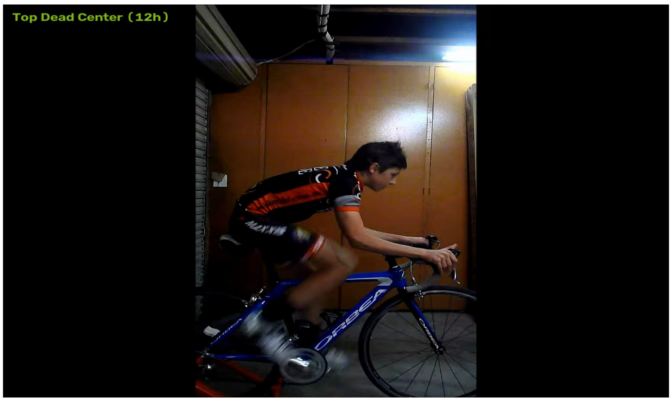
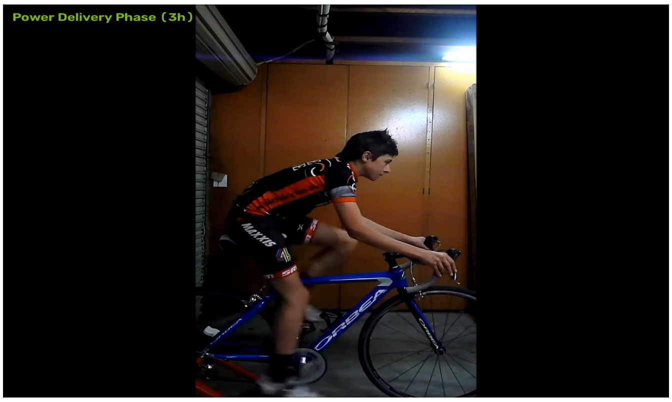
NINTAI BIOMECHANICAL MOTION DOSSIER

Dynamic Motion Capture Kinematics & Fit Evaluation

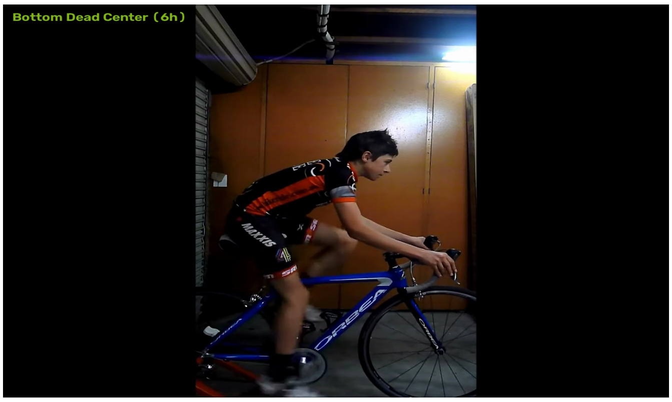
1. DYNAMIC STROKE DECOMPOSITION (4-PHASE KINEMATICS)



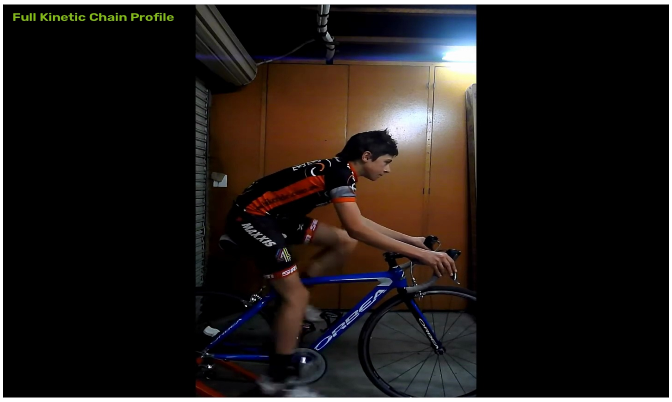
Phase 1: Top Dead Center (12h)



Phase 2: Power Delivery (3h)



Phase 3: Bottom Dead Center (6h)



Phase 4: Full Kinetic Chain Profile

2. OBSERVED BIOMECHANICAL ENVELOPES

Kinematic Metric	Observed Value	Reference Range	Status
Knee Extension (BDC 6h)	145.5 deg	140 - 150 deg	Optimal
Knee Flexion (TDC 12h)	61.8 deg	68 - 75 deg	Low / Closed
Closed Hip Angle (TDC 12h)	46.0 deg	45 - 55 deg	Optimal
Torso Incline to Horizontal	33.1 deg	40 - 50 deg	Low / Closed
Shoulder / Cockpit Angle	76.0 deg	85 - 95 deg	Low / Closed
Dynamic Ankling at BDC	91.6 deg	90 - 105 deg	Optimal

3. DIAGNOSTIC CONSULTATION & PRESCRIPTION

Clinical Biomechanical Evaluation Report

Overall Fit Alignment Score: `100% / 100%`

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1. Primary Kinematic Envelopes
- **Knee Extension at BDC (6 o'clock):** `145.5°` *(Reference: 140°?150°)*
- **Knee Flexion at TDC (12 o'clock):** `61.8°` *(Reference: 68°?75°)*
- **Closed Hip Angle at TDC:** `46.0°` *(Reference: 45°?55°)*
- **Torso Incline to Horizontal:** `33.1°` *(Reference: 40°?50°)*
- **Shoulder / Cockpit Angle:** `76.0°` *(Reference: 85°?95°)*
- **Dynamic Ankling at BDC:** `91.6°` *(Reference: 90°?105°)*

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2. Prioritized Action Plan (Step-by-Step)
1. **Primary Saddle Adjustment:**
   - **Saddle Height:** Optimal (145.5° within target 140°?150°).
2. **Cockpit & Reach Geometry:**
   - **Hip Angle:** Optimal (46.0° clearance at TDC 12h).
3. **Foot & Cleat Interface:**
   - **Dynamic Ankling:** Neutral (91.6° at BDC 6h).

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3. Overuse Injury Risk Assessment
- **Patellofemoral Tendon Load:** Low risk
- **Hamstring / Biceps Femoris Strain:** Low risk
- **Lumbar Spine Fatigue:** Within normal bounds

Note: Make adjustments in small increments (3?5mm) and perform a dynamic test ride between each change.